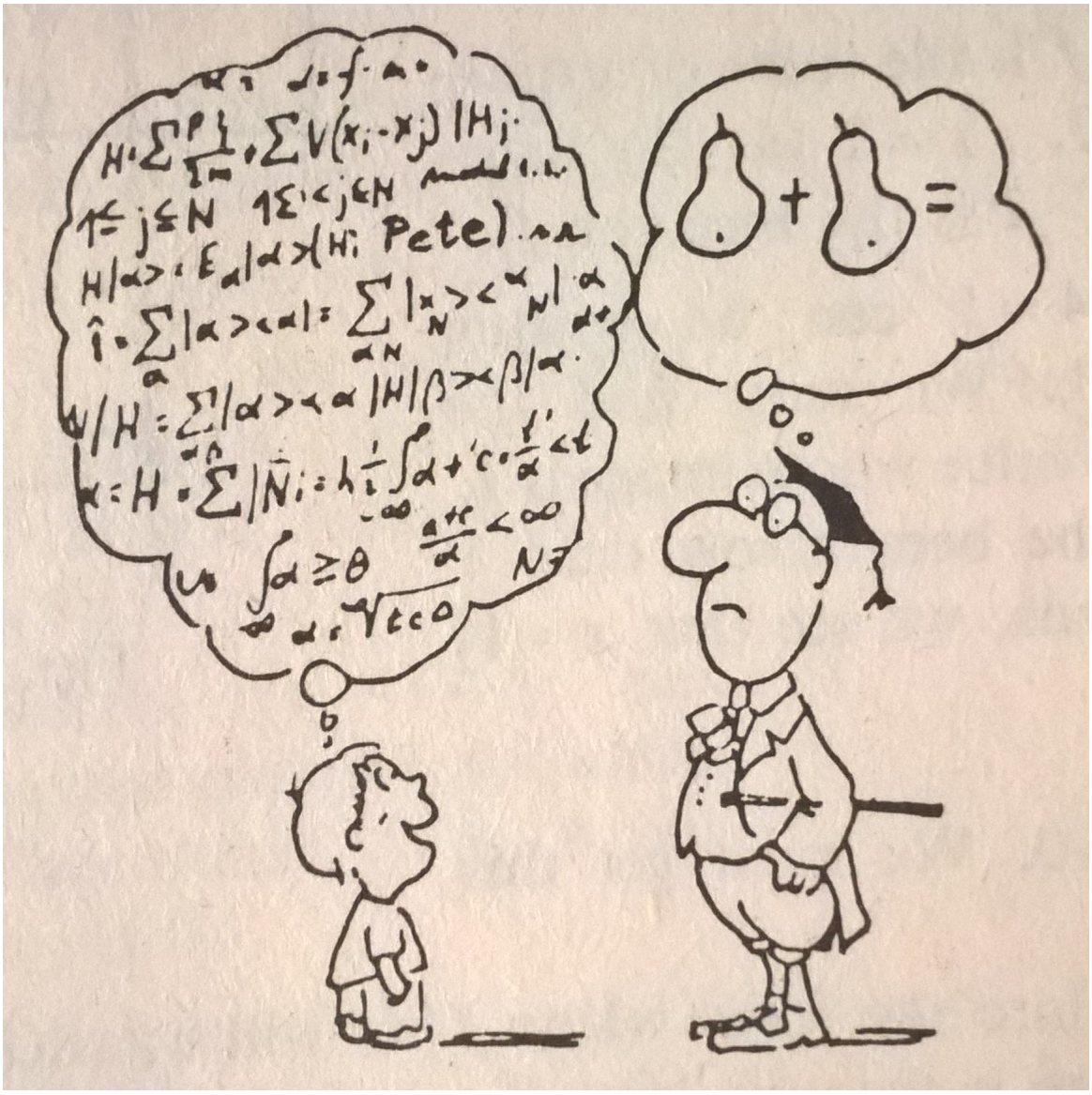
MATH BASICS


Pete

1 Definition

1.1 Exponents

1.

$$a^n \;=\; a \times a \times a \times a \;....\; n \quad factors \qquad (n \in \mathbb{N},\; a \in \mathbb{R})$$

2.

$$a^{-m} \;=\; \frac{1}{a^m} \qquad (m \in \mathbb{Z}^+,\; a \in \mathbb{R},\; a \neq 0)$$

$$and:$$

$$\frac{1}{a^{-m}} \;=\; a^m$$

3.

$$a^0 = 1 \qquad (a \in \mathbb{R},\; a \neq 0)$$

1.2 Rational Exponents:

1.

$$\sqrt[n]{a} \;=\; r \qquad (a \;>\; 0,\; n \in \mathbb{N},\; n \geqslant 2,\; r \;>\; 0),\; \Longleftrightarrow\; r^n \;=\; a$$

2.

$$a^{\frac{1}{n}} \;=\; \sqrt[n]{a}; \qquad (a \;>\; 0,\; n \;\geqslant 2,\; n \in \mathbb{N})$$

3.

$$a^{\frac{-1}{n}} \;=\; \sqrt[n]{a^{-1}}; \qquad (a \;>\; 0,\; n \;> 0,\; n \in \mathbb{N})$$

4.

$$a^{\frac{m}{n}} \;=\; \sqrt[n]{a^m}; \qquad (a \;>\; 0;\; m,\; n \in \mathbb{Z},\; n \geqslant 2)$$

2 Law

2.1 Exponents

1.

$$a^m \times a^n = a^{m+n} \qquad (m,\ n \in \mathbb{N})$$

$$a^m \times a^n = a^{m+n} \qquad (m,\ n \in \mathbb{Z};\ a \neq 0,\ if \quad m \quad or \quad n < 0)$$

2.

$$\frac{a^m}{a^n} = a^{m-n} \qquad (m,\ n \in \mathbb{Z};\ a \in \mathbb{R};\ a \neq 0)$$

3.

$$(ab)^m = a^m\, b^m \qquad (m \in \mathbb{Z})$$

4.

$$(a^m)^n = a^{mn} \qquad (m,\ n \in \mathbb{Z})$$

2.2 Rational Exponents

1.

$$a^r \times a^t = a^{r+t} \qquad (a > 0; \quad r,\ t \in \mathbb{Q})$$

2.

$$\frac{a^r}{a^t} = a^{r-t} \qquad (a > 0; \quad r,\ t \in \mathbb{Q})$$

3.

$$(a^t)^r = a^{tr} \qquad (a > 0, \quad t,\ r \in \mathbb{Q})$$

4.

$$(ab)^t = a^t\, b^t\ ; \quad \left(\frac{a}{b}\right)^t = \frac{a^t}{b^t}\ ; \qquad (a,\ b > 0, \quad t \in \mathbb{Q})$$

$$and:$$

$$a^t\, b^t = (ab)^t \qquad and \qquad \frac{a^t}{b^t} = \left(\frac{a}{b}\right)^t$$

2.3 Distributive law

$$a(b + c) = ab + ac$$

$$\begin{aligned}(a + b)(c + d) &= (a + b)c + (a + b)d \\ &= ac + bc + ad + bd\end{aligned}$$

$$A^2 - B^2 = (A - B)(A + B)$$

2.4 Commutative law

$$ab = ba$$

3 Properties

3.1 Addition

$$0 + a = a$$

$$\frac{a}{b} + \frac{c}{b} = \frac{a + c}{b} \qquad (b \neq 0)$$

$$\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b} \qquad (b \neq 0)$$

3.2 Multiplication

$$0 \times a = 0$$

$$\frac{0}{a} = 0 \times \frac{1}{a} = 0$$

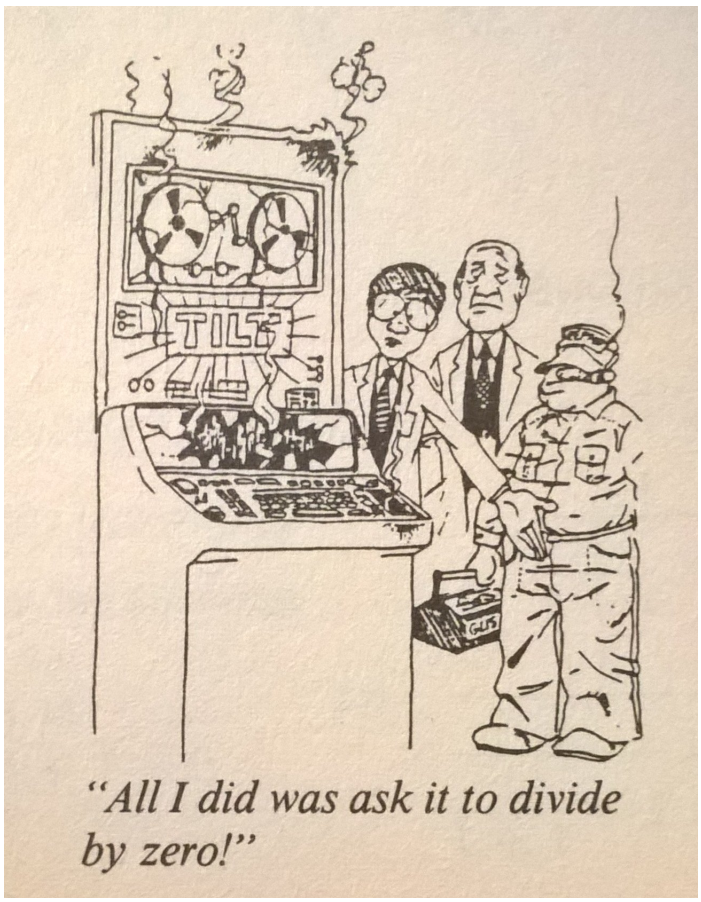
$$1 \times a = a$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \qquad (b \neq 0;\ d \neq 0)$$

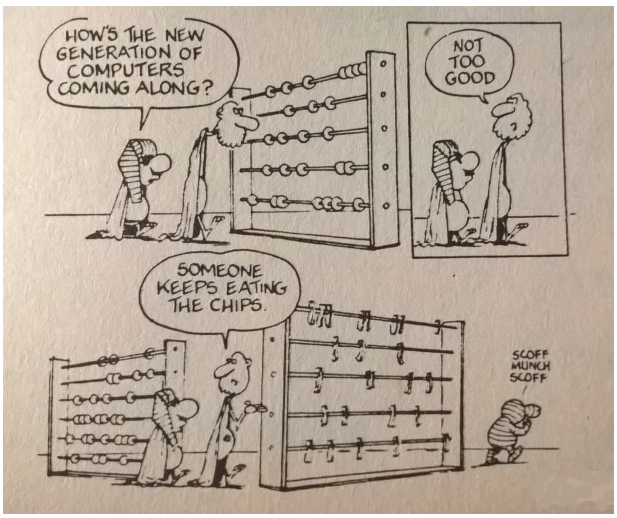
3.3 Division

$$\frac{a}{0} = undefined$$

$$\frac{p}{q} \div \frac{r}{s} = \frac{p}{q} \times \frac{s}{r} = \frac{ps}{qr} \qquad (q \neq 0;\ r \neq 0;\ s \neq 0)$$



"All I did was ask it to divide by zero!"



4 Examples

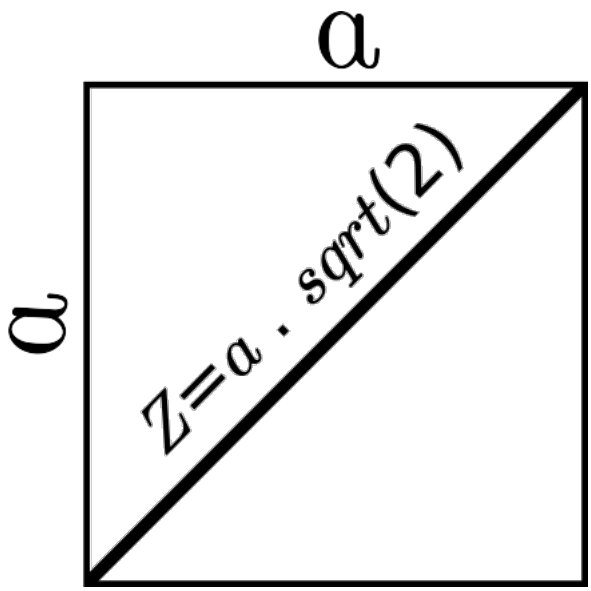
$$\sqrt{a^2} = a \qquad (a > 0) \qquad\qquad \sqrt{a^n} = a^{\frac{n}{2}}$$

$$\sqrt{\frac{1}{a}} = \frac{1}{\sqrt{a}}$$

$$a^{\frac{m}{1}} = a^m \qquad\qquad \frac{a^m}{a^n} = a^{m-n}$$

$$\frac{a \angle\alpha^\circ \; b \angle\beta^\circ}{c \angle\gamma^\circ} = \frac{a \times b}{c} \quad \angle\left(\alpha^\circ + \beta^\circ - \gamma^\circ\right)$$

$$If \quad a \,.\, b = 0, \quad then \quad a = 0 \quad or \quad b = 0$$



Square Property

$$Z = \sqrt{a^2 \ + \ a^2}$$
$$Z = \sqrt{2 \ a^2}$$
$$Z = \sqrt{2} \quad \sqrt{a^2}$$
$$Z = \sqrt{2} \quad a$$

$$a \angle\alpha \ + \ b \angle\beta \ = \ \sqrt{(a \ \sin\alpha \ + \ b \ \sin\beta)^2 \ + \ (a \ \cos\alpha \ + \ b \ \cos\beta)^2}$$

and

$$\angle \arctan(\frac{(a \ \sin\alpha \ + \ b \ \sin\beta)}{(a \ \cos\alpha \ + \ b \ \cos\beta)})$$

$$\lim_{s\to 0} GH(s) = \lim_{s\to 0} \quad 7 \ \frac{4+3s}{5+2s+6s^2}$$

$$= \lim_{s\to 0} \quad 7 \ \frac{4(1+\frac{3}{4}s)}{5(1+\frac{2}{5}s+\frac{6}{5}s^2)}$$

$$= 7 \ \frac{4}{5}$$

$$y'(x) = \mathrm{e}^{4x+5}$$
$$= 4 \ \mathrm{e}^{4x+5}$$

Derivatives made easy.

$$y = \cos^3(\operatorname{sen}(x^2+x))$$

$$y' = x^{3\,\prime}\Big|_{x=\cos(\operatorname{sen}(x^2+x))} \cdot \cos(x)'\Big|_{x=\operatorname{sen}(x^2+x)} \cdot \operatorname{sen}(x)'\Big|_{x=x^2+x} \cdot (x^2+x)'$$

$$= 3x^2\Big|_{x=\cos(\operatorname{sen}(x^2+x))} \cdot (-1)\cdot\operatorname{sen}(x)\Big|_{x=\operatorname{sen}(x^2+x)} \cdot \cos(x)\Big|_{x=x^2+x} \cdot (2x+1)$$

$$= 3\cos^2(\operatorname{sen}(x^2+x)) \cdot (-1) \cdot \operatorname{sen}(\operatorname{sen}(x^2+x)) \cdot \cos(x^2+x) \cdot (2x+1)$$

$$= -(6x+3) \cdot \cos^2(\operatorname{sen}(x^2+x)) \cdot \operatorname{sen}(\operatorname{sen}(x^2+x)) \cdot \cos(x^2+x)$$

$$y = \frac{(x+3)^3}{x^{3/2}}$$

$$y' = \frac{(x+1)^{3\,\prime} \cdot x^{3/2} - x^{3/2\,\prime} \cdot (x+1)^3}{x^3}$$

$$= \frac{3(x+1)^2 \cdot x^{3/2} - \frac{3}{2} \cdot x^{1/2} \cdot (x+1)^3}{x^3} \qquad \text{HCF}$$

$$= \frac{\left[3(x+1)^2 \cdot x - \frac{3}{2}(x+1)^3\right] x^{\frac{1}{2}}}{x^3}$$

$$= \frac{3(x+1)^2 \cdot x - \frac{3}{2}(x+1)^3}{x^{5/2}} \qquad \text{HCF}$$

$$= \frac{(x+1)^2 \cdot \left(3x - \frac{3}{2}(x+1)\right)}{x^{5/2}}$$

$$= \frac{(x+1)^2 \cdot \left(3x - \frac{3}{2}x - \frac{3}{2}\right)}{x^{5/2}}$$

$$= \frac{(x+1)^2 \cdot \left(\frac{3}{2}x - \frac{3}{2}\right)}{x^{5/2}}$$

$$= \frac{(x+1)^2 \cdot \frac{3}{2}(x-1)}{x^{5/2}}$$

$$= \frac{3}{2} \cdot \frac{(x+1)^2 \cdot (x-1)}{x^{5/2}}$$

c.a) $\frac{3}{2} - \frac{2}{2} = \frac{1}{2}$

$x^{\frac{3}{2}} = x + x^{\frac{1}{2}}$

$\frac{1}{2} - \frac{6}{2} = -\frac{5}{2}$

$\frac{6}{2} - \frac{3}{2} = \frac{3}{2}$

$$\int x \;=\; \frac{x^2}{2}$$

$$\int_{-1}^{1} \sqrt{\frac{1+x}{1-x}} = \pi$$

$$e = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + ... + \frac{1}{n!} + ...$$

5 Methods

HCF - highest common factor (pôr variável em evidência)

Factorisation

LCD or LCM - Lowest common denominator or lowest common multiple

Capítulo 1

Definições

Definição 1 *Capacitância*

$$
\begin{aligned}
Q_c(t) =& \int^t i(t) \quad dt \\
=& Q_c(0^-) + \int_{0^-}^t i(t) \quad dt \\
V_c(t) =& \frac{Q_c(t)}{C} \\
=& \frac{1}{C} \quad \int^t i_c(t) \quad dt \\
=& \frac{Q_c(0^-)}{C} + \frac{1}{C} \quad \int_0^t i_c(t) \quad dt \\
=& V(0^-) + \frac{1}{C} \quad \int_0^t i_c(t) \quad dt \\
i_c(t) =& C \quad \frac{dV_c(t)}{dt} \\
W =& \frac{1}{2}\, C\, V^2
\end{aligned}
$$

Definição 2 *Indutância*

$$
\begin{aligned}
\psi_L(t) =& \int^t V_L(t) \quad dt \\
=& \psi_L(0^-) + \int_{0^-}^t V_L(t) \quad dt \\
V_L(t) =& L \quad \frac{di_L(t)}{dt} \\
i_L(t) =& \frac{\psi_L(t)}{L} \\
=& \frac{1}{L} \quad \int^t V_L(t) \quad dt \\
=& \frac{\psi_L(0^-)}{L} + \frac{1}{L} \quad \int_0^t V_L(t) \quad dt \\
=& i_L(0^-) + \frac{1}{L} \quad \int_0^t V_L(t) \quad dt \\
W =& \frac{1}{2}\, L\, i^2
\end{aligned}
$$

Definição 3 *Resistência*

$$
\begin{aligned}
V_R(t) &= R \quad i_R(t) \\
i_R(t) &= \frac{V_R(t)}{R} \\
P &= Ri^2 \\
P &= \frac{U^2}{R} \\
W &= P\, \Delta t
\end{aligned}
$$

Definição 4 *Valor Médio*

$$X_{av} = \frac{1}{T} \int_0^T X(t)dt$$

Definição 5 *Valor Eficaz*

$$X_{ef} = \sqrt{\frac{1}{T} \int_0^T X^2(t)dt}$$